

P4: Structuring Code

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Overview

As the logic in your program grows, you will need better ways to organise your code. Learn how to structure your code to better organise the logic and your thoughts using functions and procedures.

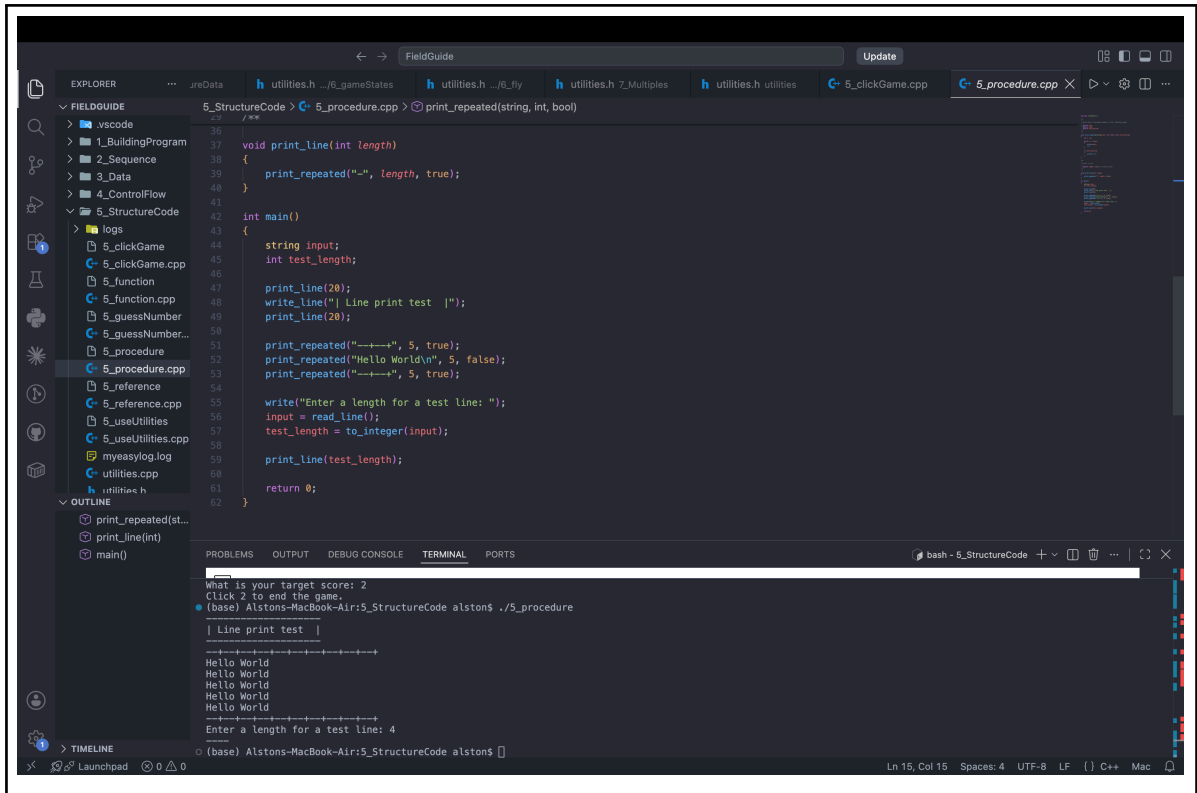
Tip: Abstract thinking

You will now be starting to explore tools that lift the level of abstraction you can work with. Now rather than solving a specific problem, you can produce code constructs that solve more general versions of these problems - meaning they can be used in a wider range of

Do the following:

- Everything you need is in the [Structuring Code](#) chapter of our [Programmer's Field Guide](#). The [Intro Tours](#) walk through the concepts, and should give you a good start.
- Add a screenshot showing you were able to create a simple procedure with parameters. You can use the [explore procedure tour](#).

```
FieldGuide
5_StructureCode > utilities.h 6_StructureData utilities.cpp 6_StructureData utilities.h
5_StructureCode > 5_procedure.cpp > print_repeated(string, int, bool)
1  #include "splashkit.h"
2
3  /**
4   *
5   * print text to terminal a number of lines, allowing repeat
6   *
7   * @param text
8   * @param times
9   * @param with_newline
10  *
11  */
12
13 void print_repeated(string text, int times, bool with_newline)
14 {
15     int i = 0;
16
17     while (i < times)
18     {
19         write(text);
20         i++;
21     }
22
23     if (with_newline)
24     {
25         write('\n');
26     }
27
28
29 /**
30 *
31 * print a line
32 *
33 * @param length length of line to print
34 *
35 */
36
37 void print_line(int length)
38 {
39     print_repeated("-", length, true);
40 }
41
42 int main()
43 {
44     string input;
45     int test_length;
46
47     print_line(20);
48     write_line("| Line print test |");
49     print_line(20);
50
51     print_repeated("---+", 5, true);
52     print_repeated("Hello World\n", 5, false);
53     print_repeated("---+", 5, true);
54
55     write("Enter a length for a test line: ");
56     input = read_line();
57     test_length = to_integer(input);
58
59     print_line(test_length);
60
61     return 0;
62 }
```



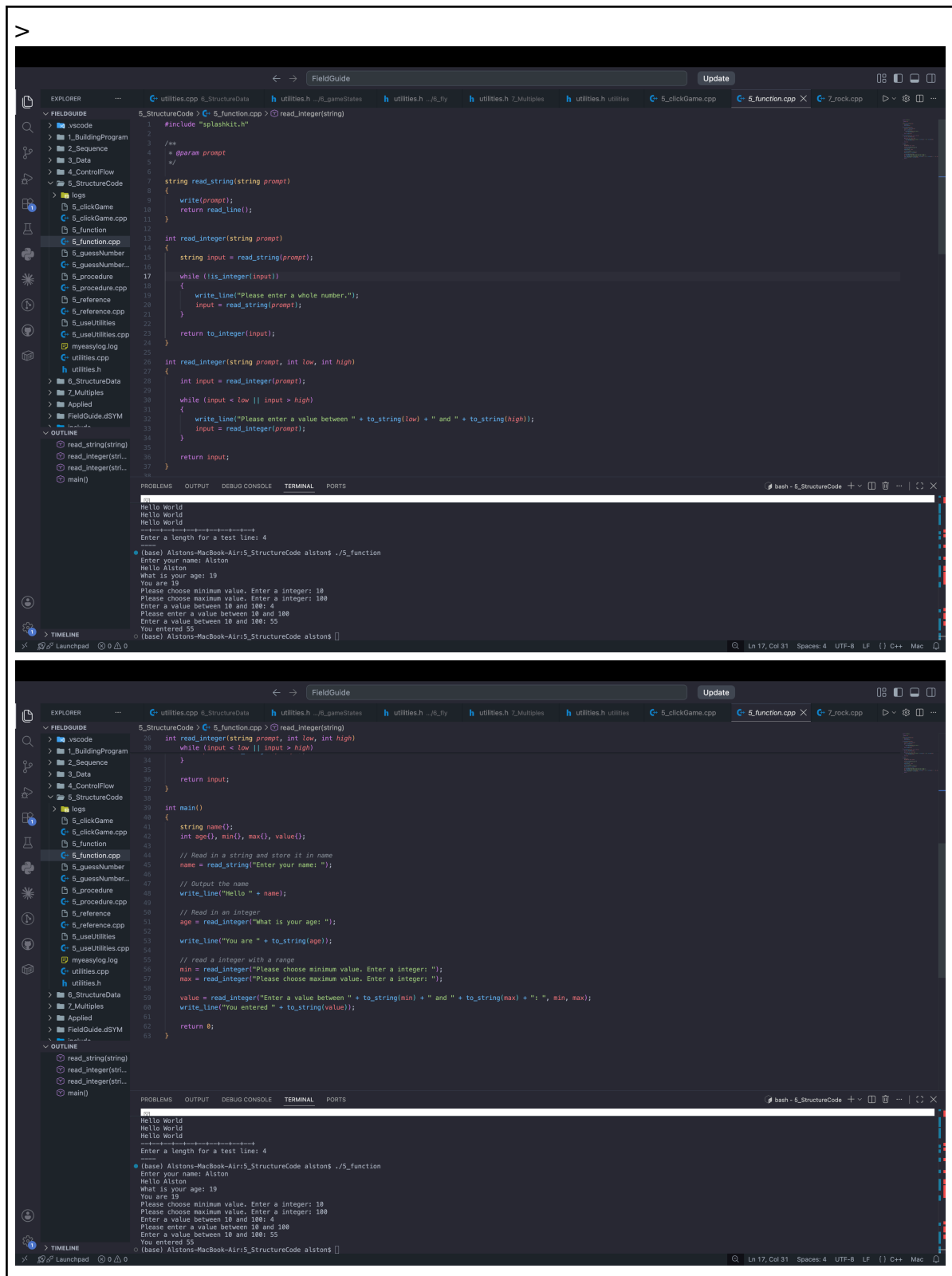
The screenshot shows the Visual Studio Code editor with a C++ project named '5_StructureCode'. The file '5_procedure.cpp' is open, showing a function `print_line` that takes an integer `length` and a boolean `bool` parameter. The function prints a line of dashes followed by the word 'Hello World'. The `main` function calls `print_line` with `20` and `true`, then prompts the user for a test line length. The terminal shows the output of the program, including the printed line and the user's input of `4`.

```
27 // **
28
29 void print_line(int length)
30 {
31     print_repeated("-", length, true);
32 }
33
34 int main()
35 {
36     string input;
37     int test_length;
38
39     print_line(20);
40     write_line("| Line print test |");
41     print_line(20);
42
43     print_repeated("----+", 5, true);
44     print_repeated("Hello World\n", 5, false);
45     print_repeated("----+", 5, true);
46
47     write("Enter a length for a test line: ");
48     input = read_line();
49     test_length = to_integer(input);
50
51     print_line(test_length);
52
53     return 0;
54 }
```

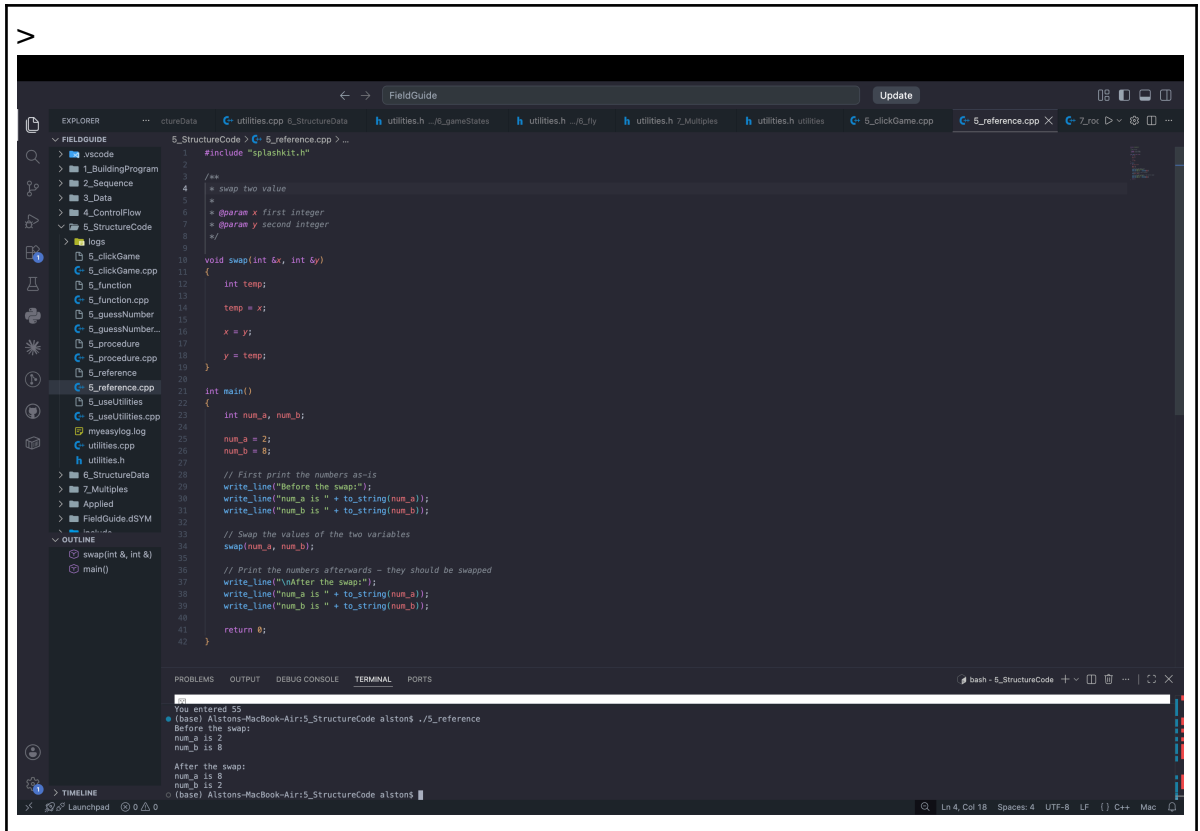
Terminal Output:

```
What is your target score: 2
Click 2 to end the game.
(base) Alstons-MacBook-Air:5_StructureCode alston$ ./5_procedure
| Line print test |
-----+
Hello World
Hello World
Hello World
Hello World
Hello World
-----+
Enter a length for a test line: 4
```

- Add a screenshot showing you were able to create a simple function with parameters. You can use the [explore functions tour](#).



- Add a screenshot showing you were able to use reference parameters. You can use the [explore pass by reference tour](#).

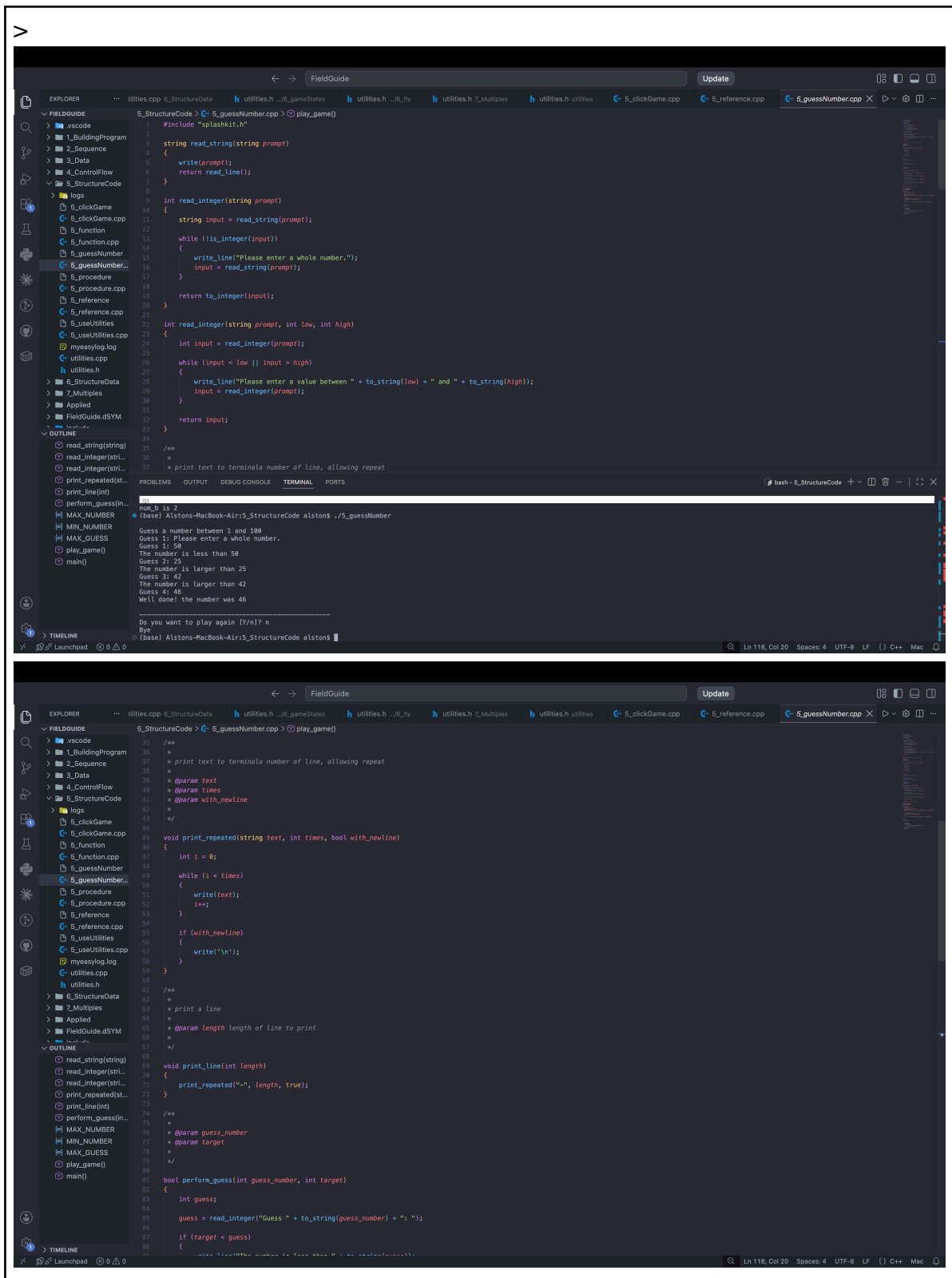


```
>
EXPLORER 5_StructureCode > 5_reference.cpp > ...
  1 #include "splashkit.h"
  2
  3 /**
  4  * swap two value
  5  * @param x first integer
  6  * @param y second integer
  7  */
  8 void swap(int &x, int &y)
  9 {
 10     int temp;
 11     temp = x;
 12     x = y;
 13     y = temp;
 14 }
 15
 16 int main()
 17 {
 18     int num_a, num_b;
 19     num_a = 2;
 20     num_b = 8;
 21
 22     // First print the numbers as-is
 23     write_line("Before the swap:");
 24     write_line("num_a is " + to_string(num_a));
 25     write_line("num_b is " + to_string(num_b));
 26
 27     // Swap the values of the two variables
 28     swap(num_a, num_b);
 29
 30     // Print the numbers afterwards - they should be swapped
 31     write_line("After the swap:");
 32     write_line("num_a is " + to_string(num_a));
 33     write_line("num_b is " + to_string(num_b));
 34
 35     return 0;
 36 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
bash - 5_StructureCode
You entered 55
(base) Alstons-MacBook-Air:5_StructureCode alstons ./5_reference
Before the swap:
num_a is 2
num_b is 8
After the swap:
num_a is 8
num_b is 2
(base) Alstons-MacBook-Air:5_StructureCode alstons
```

- Add a screenshot showing you were able to put these together in a small program. You can use either the [guess that number tour](#) or the [change calculator tour](#).



The screenshot displays the Visual Studio Code editor interface. The Explorer view on the left shows a project structure with files like `5_StructureCode`, `5_guessNumber.cpp`, and `5_clickGame.cpp`. The Source Control view in the middle shows a list of files. The Output view at the bottom shows the execution of the program, displaying prompts and user input.

```

1 #include "splashkit.h"
2
3 string read_string(string prompt)
4 {
5     write(prompt);
6     return read_line();
7 }
8
9 int read_integer(string prompt)
10 {
11     string input = read_string(prompt);
12     while (!is_integer(input))
13     {
14         write_line("Please enter a whole number.");
15         input = read_string(prompt);
16     }
17     return to_integer(input);
18 }
19
20 int read_integer(string prompt, int low, int high)
21 {
22     int input = read_integer(prompt);
23     while (input < low || input > high)
24     {
25         write_line("Please enter a value between " + to_string(low) + " and " + to_string(high));
26         input = read_integer(prompt);
27     }
28     return input;
29 }
30
31 //**
32 * print text to terminal a number of line, allowing repeat
33 *
34 * @param text
35 * @param times
36 * @param with_newline
37 *
38 void print_repeated(string text, int times, bool with_newline)
39 {
40     int i = 0;
41     while (i < times)
42     {
43         write(text);
44         i++;
45     }
46     if (with_newline)
47     {
48         write('\n');
49     }
50 }
51
52 //**
53 * print a line
54 *
55 * @param length length of line to print
56 *
57 void print_line(int length)
58 {
59     print_repeated("-", length, true);
60 }
61
62 //**
63 *
64 * @param guess_number
65 * @param target
66 *
67 bool perform_guess(int guess_number, int target)
68 {
69     int guess;
70     guess = read_integer("Guess " + to_string(guess_number) + ": ");
71     if (target < guess)
72     {
73         write_line("Too high!");
74     }
75     else if (target > guess)
76     {
77         write_line("Too low!");
78     }
79     else
80     {
81         write_line("Correct!");
82     }
83 }
84
85 int main()
86 {
87     perform_guess(1, 100);
88     return 0;
89 }

```

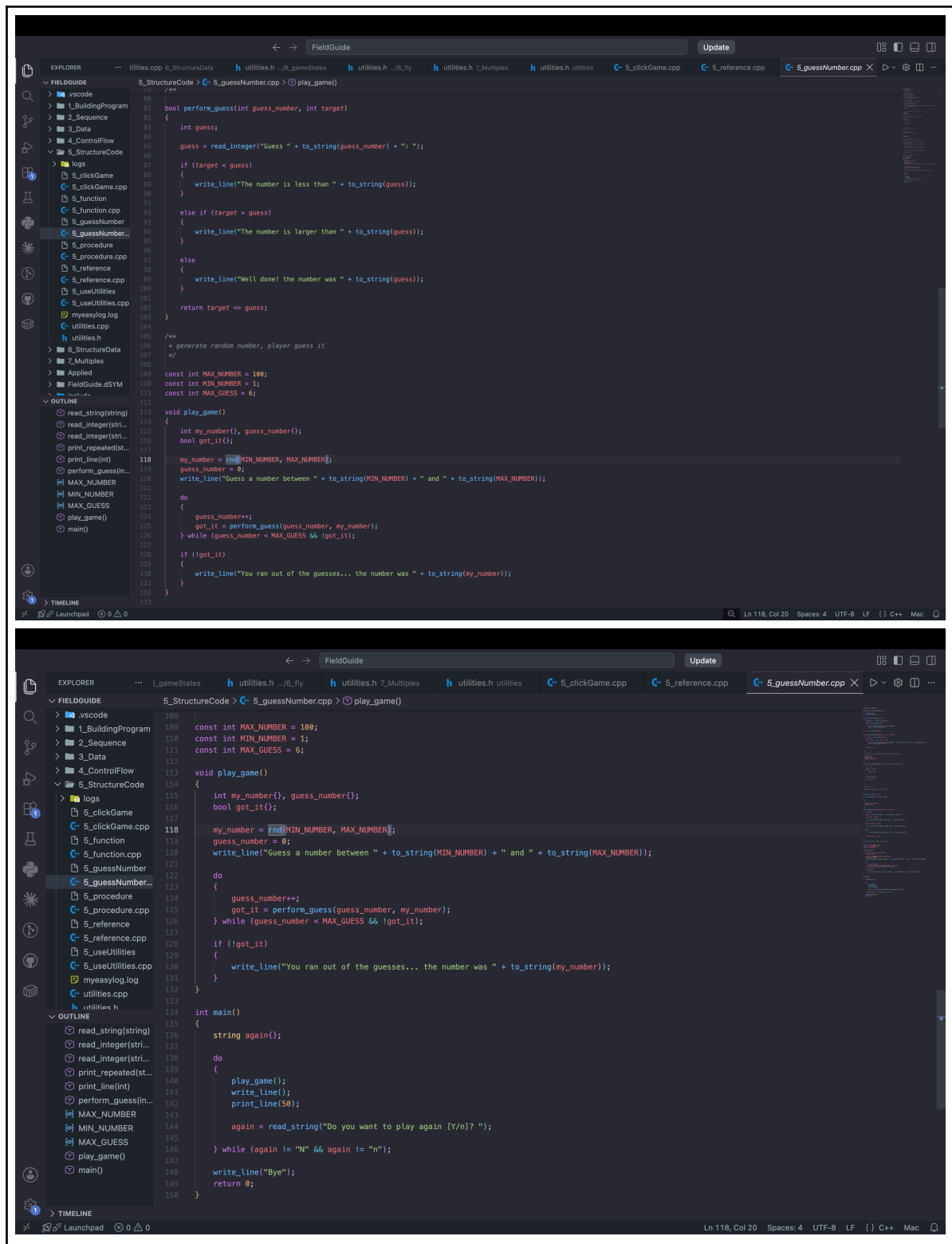
Output:

```

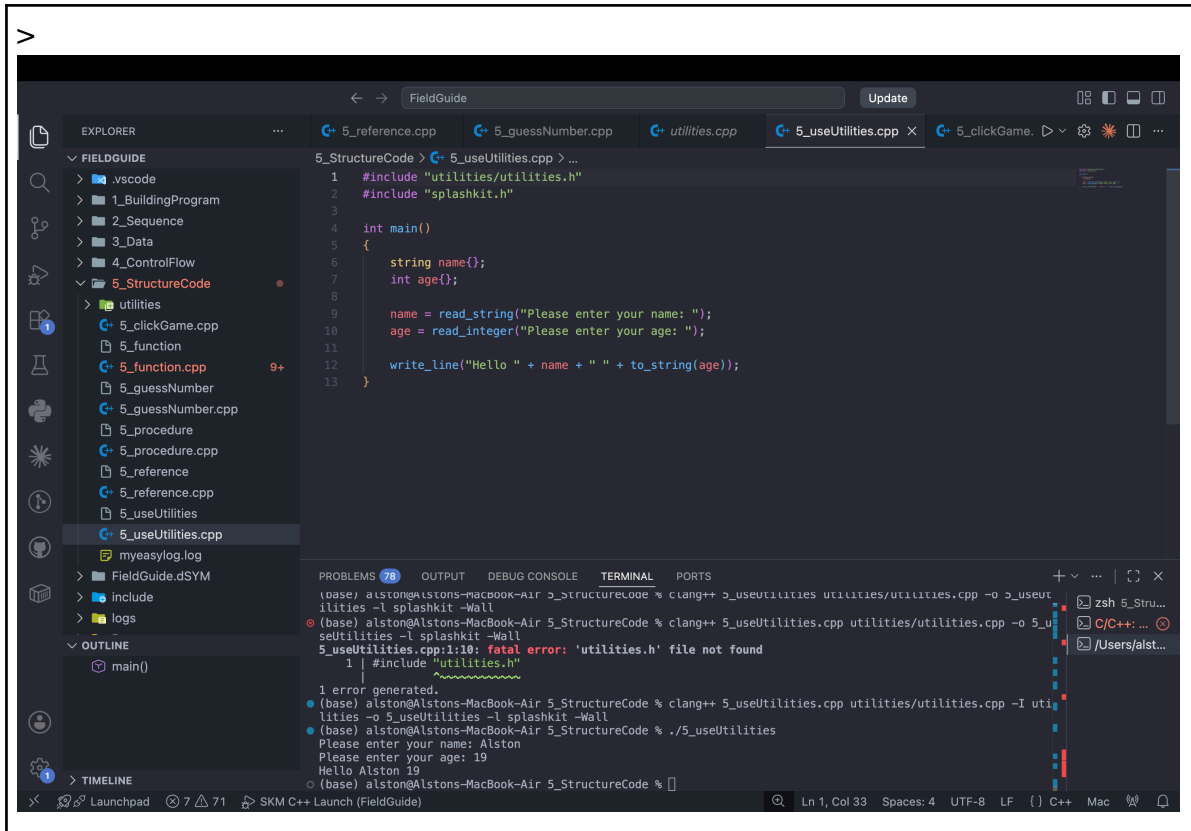
(base) Alstons-MacBook-Air:5_StructureCode alstons$ ./5_guessNumber
Guess a number between 1 and 100
Guess 1: Please enter a whole number.
Guess 1: 50
The number is less than 50
Guess 2: 25
The number is larger than 25
Guess 3: 42
The number is larger than 42
Guess 4: 46
Well done! the number was 46

Do you want to play again (Y/N)? n
Bye

```



- Add a screenshot showing you were able to create shared utilities. You can use the [shared utilities tour](#).



```

1 #include "utilities/utilities.h"
2 #include "splashkit.h"
3
4 int main()
5 {
6     string name{};
7     int age{};
8
9     name = read_string("Please enter your name: ");
10    age = read_integer("Please enter your age: ");
11
12    write_line("Hello " + name + " " + to_string(age));
13 }

```

PROBLEMS (76) OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

(base) alston@Alstons-MacBook-Air 5_StructureCode % clang++ 5_useUtilities.cpp utilities/utilities.cpp -o 5_useUtilities -l splashkit -Wall
(base) alston@Alstons-MacBook-Air 5_StructureCode % clang++ 5_useUtilities.cpp utilities/utilities.cpp -o 5_useUtilities -l splashkit -Wall
5_useUtilities.cpp:1:10: fatal error: 'utilities.h' file not found
1 | #include "utilities.h"
  |          ^
1 error generated.
(base) alston@Alstons-MacBook-Air 5_StructureCode % clang++ 5_useUtilities.cpp utilities/utilities.cpp -I utilities -o 5_useUtilities -l splashkit -Wall
(base) alston@Alstons-MacBook-Air 5_StructureCode % ./5_useUtilities
Please enter your name: Alston
Please enter your age: 19
Hello Alston 19
(base) alston@Alstons-MacBook-Air 5_StructureCode %

```

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- Complete one of the [Test Your Capabilities](#) tasks from the chapter to see that you can now put these tools together to achieve something yourself.

> I choose click game, first I refer to control flow program so then can have some idea... I set all the variable use in full time and write some word, and also use random to set the initial position of target. I want to do a moving target so is more interesting, so I need to think more.

In do while, I separate my work to some part: renew target coordinate, change direction if target touch boundary, check and count when hit, and last is draw target and progressing bar.

I use speed to change target(x, y)

And I tried many time to change the direction because it include many boolean logic.

In check hit, I use hit_count++ to renew the count.

Drawing is more trouble for me because need to think about coordinate of each shapes and point! I use all variables to replace the number inside the fill_shape() function so I can change it conveniently.

Last, I use do while when count less than target score and !quit to provide to ways to quit the program.

I resubmit it by enclosed all thing into function in utilities.

>

FieldGuide

Update

EXPLORER

FieldGuide

5_StructureCode > 5_clickGame.cpp > update_game(int &int &, int &, int &, int &, int &, int &)

```

1 #include "splashkit.h"
2 #include "utilities.h"
3
4 const int CIRCLE_RADIUS = 100;
5
6 // Check if the target has been hit, when hit, return true
7 bool target_hit(double x, double y, double radius)
8 {
9     return point_in_circle(mouse_x(), mouse_y(), x, y, radius) && mouse_clicked(LEFT_BUTTON);
10 }
11
12 // Give target random position
13 void random_target_position(int &target_x, int &target_y, int window_x, int window_y)
14 {
15     target_x = rnd(CIRCLE_RADIUS, window_x - CIRCLE_RADIUS);
16     target_y = rnd(CIRCLE_RADIUS, window_y - CIRCLE_RADIUS);
17 }
18
19 // change direction of velocity when hit boundary
20 void change_direction(int target_x, int target_y, int window_x, int window_y, int &speed_x, int &speed_y)
21 {
22     if ((target_x > (window_x - CIRCLE_RADIUS) && speed_x > 0) || (target_x < CIRCLE_RADIUS && speed_x < 0))
23     {
24         speed_x = -speed_x;
25     }
26     if ((target_y > (window_y - CIRCLE_RADIUS) && speed_y > 0) || (target_y < (CIRCLE_RADIUS + 100) && speed_y < 0))
27     {
28         speed_y = -speed_y;
29     }
30 }
31
32 // Draw a progress bar
33 void draw_progress(double bar_x, double bar_y, double bar_width, double bar_height, double percent)
34 {
35     // fill the black background of the progress bar
36     fill_rectangle(COLOR_BLACK, bar_x, bar_y, bar_width, bar_height);
37
38     // setting the data of the green progress bar
39     double progress_length = bar_width * 0.999;
40     double progress_height = bar_height * 0.96;
41     double progress_x = bar_x + (bar_width - progress_length) / 2;
42     double progress_y = bar_y + (bar_height - progress_height) / 2;
43
44     // fill line green progress bar with it ratio
45     fill_rectangle(COLOR_LINE_GREEN, progress_x, progress_y, progress_length * percent, progress_height);
46 }
47
48 // renew target coordinate
49 void renew_target_coordinate(int &target_x, int &target_y, int &speed_x, int &speed_y, int window_x, int window_y)
50 {
51     // renew the position with current speed (x, y)
52     // moving toward direction x and y according to the speed
53     target_x += speed_x;
54     target_y += speed_y;
55 }

```

Timeline

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FieldGuide

Update

EXPLORER

FieldGuide

5_StructureCode > 5_clickGame.cpp > update_game(int &int &, int &, int &, int &, int &, int &)

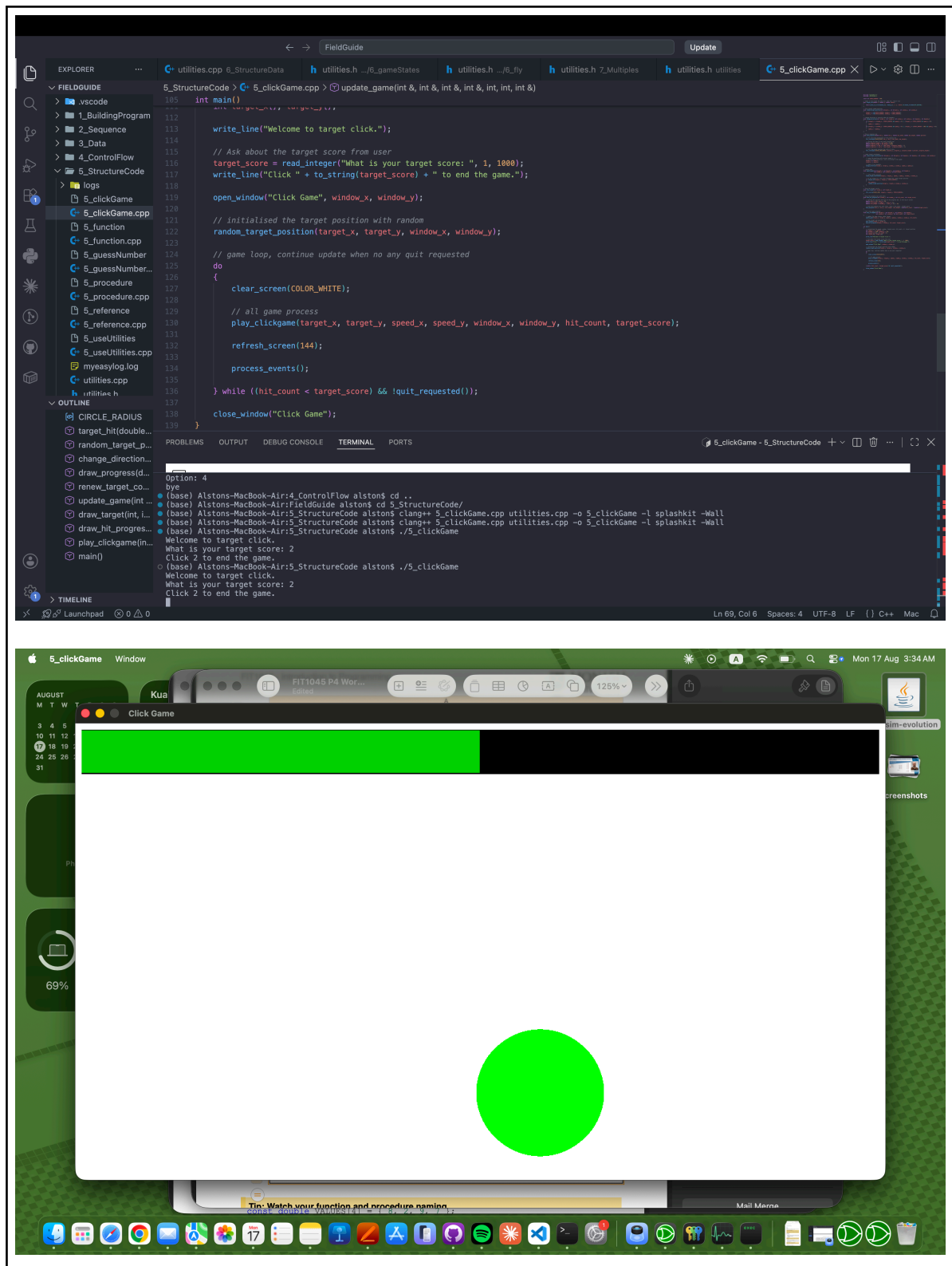
```

61 // update game
62 void update_game(int &target_x, int &target_y, int &speed_x, int &speed_y,
63                 int window_x, int window_y, int &hit_count)
64 {
65     // renew target position
66     random_target_position(target_x, target_y, speed_x, speed_y, window_x, window_y);
67
68     // if the target hit, hit_count increase one, renew target position
69     if (target_hit(target_x, target_y, CIRCLE_RADIUS))
70     {
71         hit_count++;
72         random_target_position(target_x, target_y, window_x, window_y);
73     }
74
75     // draw the target circle
76     void draw_target(int target_x, int target_y)
77     {
78         fill_circle(COLOR_LIME, target_x, target_y, CIRCLE_RADIUS);
79     }
80
81     // Draw the progress bar with ratio
82     void draw_hit_progress(int window_x, int window_y, int hit_count, int target_score)
83     {
84         // setting the position and size of the progress bar, at the top of screen
85         double rect_x = 10, rect_y = 10;
86         double rect_length = window_x - (rect_x * 2);
87         double rect_height = (window_y - (rect_y * 2)) / 10;
88
89         // draw the progress bar with ratio, ratio = hit_count / target_score
90         draw_progress(rect_x, rect_y, rect_length, rect_height, (double)hit_count / (double)target_score);
91     }
92
93     // full click game progress
94     void play_clickgame(int &target_x, int &target_y, int &speed_x, int &speed_y,
95                        int window_x, int window_y, int &hit_count, int target_score)
96     {
97         // update the game status, about target
98         update_game(target_x, target_y, speed_x, speed_y, window_x, window_y, hit_count);
99
100        // draw target and progress bar
101        draw_target(target_x, target_y);
102        draw_hit_progress(window_x, window_y, hit_count, target_score);
103    }
104
105    int main()
106    {
107        // initialized the speed, window, target_score, hit_count = 0, target position
108        int speed_x = 1, speed_y = 1;
109        int window_x = 1280, window_y = 720;
110        int target_score(), hit_count = 0;
111        int target_x(), target_y();
112
113        write_line("Welcome to target click.");
114    }

```

Timeline

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Tip: Watch your function and procedure naming

Make sure to check the [style guide](#) sections around creating functions and procedures.

Reflect:

- What was the most interesting thing you learnt through this task, and why?

> Is about the procedure very important so that we can did a complex task!

- How do functions and procedures help you approach the development of larger programs?

> Make me more easier and without repeat for my work, which can follow the rule "DRY".
If no function, we need to repeat many time for a same thing which is not efficient.

- How do parameters make your functions and procedures more useful? Give examples from your work.

> Parameters make function more flexible in each situation, with different input. For example is `read_string()` and `read_integer()` is very useful in different program. And sometime using & for swap, for example use in each (x,y) which always be changed.

- Why are global variables considered poor practice, when global constants are considered good practice?

> Because global constants cannot change it, so that can prevent bug. Global variables is always in changing may cause bug.

- Looking back, how long did it take to complete this task? What did you spend the most time on?

> 8 hours, most time in click game, I need to think many thing.
And take another 3 hour for resubmit.

Disclose:

- Explain your use of Generative AI for this task. If you didn't use it, just write "I did not use any AI for this task". If you did use it, please state which AI model you used (e.g. ChatGPT) and how.

> I used Claude. I ask about the terminal command when utilities in different file but I change it later. I also let Claude check of my program sequence in click game.

Submit:

- Export this worksheet as a PDF.
- Upload your submission to OnTrack:

- Engage with feedback and discussion to get this signed off.